

VIETNAM NATIONAL UNIVERSITY INTERNATIONAL SCHOOL



BUSINESS INNOVATION AND IT

Topic: Technological Innovation Strategy of Oracle AI Database 26ai

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TABLE OF CONTENTS

PART I: INTRODUCTION	1
1.1. Introduction to Oracle Corporation	1
1.2. Innovation products and services analyzed	1
PART II: INDUSTRY DYNAMICS OF TECHNOLOGICAL INNOVATION	1
2.1. Sources of innovation at Oracle	1
2.1.1. Required examples of staff, user, R&D, and external innovation	2
2.2. Translating creativity into innovation	3
2.2.1. How the idea is implemented commercially	4
2.3. Types and patterns of innovation at Oracle	5
2.3.1. Why the classification is mixed rather than single category	5
2.4. Technology S-curves and technology cycles	6
PART III: FORMULATING TECHNOLOGICAL INNOVATION STRATEGY	7
3.1. Product overview and project-selection logic	7
3.2. External analysis: Porter's Five Forces	8
3.3. Internal analysis: Porter's Value Chain	10
3.4. Core competencies and VRIO analysis	13
3.5. Collaboration strategy and innovation protection	14
PART IV: IMPLEMENTING TECHNOLOGICAL INNOVATION STRATEGY	15
4.1. New Product Development process for Oracle AI Database 26ai	15
4.2. Implementation team structure for Oracle AI Database 26ai	18
4.3. Product Prototype: Oracle AI Database 26ai Interface	20
PART V: CONCLUSION	24
REFERENCES	25

LIST OF FIGURES AND TABLES

Figure 1. Oracle AI Database 26ai: Innovation Process.....	3
Figure 2. Oracle AI World 2025: Key Announcements and Strategic Metrics.....	7
Figure 3. Oracle AI Database 26ai: Architecture Overview	7
Figure 4. Value Chain Analysis: Oracle AI Database 26ai (Porter, 1985).....	10
Figure 5. Stage-Gate and Agile Hybrid NPD Process.....	16
Figure 6. Publicly Evidenced Oracle Implementation Structure.....	19
Figure 7. Prototype Landing Page: Oracle AI Database 26ai Enterprise AI-Native Platform.....	21
Figure 8. Prototype Architecture View: Five-Layer AI Platform	21
Figure 9. Prototype Feature View: Enterprise AI Capabilities.....	22
Figure 10. Prototype Multicloud View: Run Anywhere Enterprise AI.....	23
Figure 11. Prototype Call-to-Action View: Start Building with Oracle AI Database 26ai.....	23
Table 1. Sources of Innovation at Oracle	2
Table 2. Creativity-to-Innovation Process Stages	5
Table 3. Types and Patterns of Innovation.....	6
Table 4. Porter's Five Forces Assessment	9
Table 5. Value Chain: Primary and Support Activities.....	10

PART I: INTRODUCTION

1.1. Introduction to Oracle Corporation

Oracle Corporation is a U.S.-based enterprise technology company founded in 1977 by Larry Ellison, Bob Miner, and Ed Oates. Starting with relational database management systems (RDBMS), Oracle gradually expanded into enterprise applications, cloud infrastructure, and AI-enabled data platforms. As of FY2025, Oracle reported revenue of \$57.4 billion and R&D spending of \$9.86 billion, numbers that reflect both its market size and its sustained investment in innovation (Oracle Corporation, 2025a). What makes Oracle especially relevant for this course is the challenge it faces: how does a large, established firm renew itself when the technology landscape shifts? Oracle's current AI push, anchored by Oracle AI Database 26ai and OCI, is a live example of exactly that question.

1.2. Innovation products and services analyzed

This report focuses on Oracle AI Database 26ai as the primary innovation case, with Oracle Cloud Infrastructure (OCI) as the supporting platform. Announced in October 2025, Oracle AI Database 26ai embeds AI directly into the database engine itself, offering AI Vector Search, autonomous management, in-database AI agents, and a built-in AI Lakehouse, all without requiring enterprises to move their data to a separate AI system (Oracle Corporation, 2025b). OCI complements this with the compute power needed to run large AI workloads at scale, including the new OCI Zettascale10 cluster delivering up to 16 zettaFLOPS of processing capacity (Oracle Corporation, 2025c). Together, they represent Oracle's attempt to position the database, not the cloud platform, as the natural home for enterprise AI.

PART II: INDUSTRY DYNAMICS OF TECHNOLOGICAL INNOVATION

2.1. Sources of innovation at Oracle

Schilling's sources-of-innovation framework emphasizes that innovation can originate from individuals, firms, users, universities, governments, nonprofit organizations, and the linkages among these actors (Schilling, 2023). Oracle's case fits this network-based view because its AI database strategy depends on both internal and external contributors.

Table 1. Sources of Innovation at Oracle

Source of innovation	Oracle example	Contribution to Oracle AI Database 26ai and OCI
Individuals	Larry Ellison, Juan Loaiza, OCI leadership	Set strategic vision for AI- native database and mission- critical cloud infrastructure.
In-house R&D	Oracle database engineering, Oracle Labs, global R&D workforce	Build core database engine, AI Vector Search, security, performance, and cloud integration.
Users / lead customers	Large enterprise customers, enterprise customer requirements, customer feedback, and industry-specific deployment needs	Reveal needs around security, data governance, multicloud deployment, and AI search performance.
Suppliers and partners	NVIDIA, AMD, OpenAI, Microsoft Azure, Google Cloud, AWS	Provide AI hardware, model ecosystem, cloud reach, and complementary infrastructure.
Universities and government-related ecosystem	Oracle Academy, public-sector cloud requirements, regulated- industry needs	Support talent development, compliance requirements, and long-term knowledge diffusion.
Acquisitions	Sun Microsystems, NetSuite, Cerner	Add capabilities and markets that are difficult to build quickly from scratch.

In short, Oracle's innovation doesn't come from a single source. The most sensitive technical work stays internal, while everything else , hardware, AI models, cloud reach, regulatory expertise , is brought in through partnerships and acquisitions. Schilling's framework describes this as a hybrid innovation network, and Oracle is a textbook example of it working at enterprise scale (Schilling, 2023).

2.1.1. Required examples of staff, user, R&D, and external innovation

Breaking it down more concretely, Oracle's innovation comes from four groups. First, internal engineers, spread across Oracle Labs, the database engineering team, and OCI infrastructure, build the proprietary capabilities that rivals can't copy overnight: AI Vector Search, Autonomous

Tuning, the core RDBMS engine. These aren't new ideas; they're the product of decades of accumulated technical knowledge and continuous internal experimentation.

Second, enterprise customers also shape Oracle's innovation direction by creating clear demand for secure, governed, and scalable AI-data solutions. Banks, hospitals, telecom companies, and public-sector organizations may not directly invent Oracle's features, but their operational requirements influence what Oracle needs to prioritize: stronger data governance, compliance support, AI search accuracy, multicloud deployment, and high availability. In this sense, Oracle AI Database 26ai can be understood as a response to enterprise customer requirements rather than only an internal technical invention. This also matches von Hippel's view that users often reveal important needs before firms fully commercialize a solution (von Hippel, 2005).

Third, the R&D function itself, funded at roughly \$9.86 billion per year, drives the longer-horizon exploration that feeds future products (Oracle Corporation, 2025a). Oracle Labs, in particular, researches areas like distributed computing, AI inference, and database query optimization well before they reach mainstream commercialization. Fourth, external partners fill the gaps: NVIDIA provides the GPU infrastructure; OpenAI brings the model ecosystem through Stargate; and Microsoft, Google, and AWS turn competitive tension into distribution opportunity through the Database@Cloud partnerships (OpenAI, 2025). Each group contributes what the others can't easily replicate on their own, and Oracle AI Database 26ai is the result of all four working together.

2.2. Translating creativity into innovation

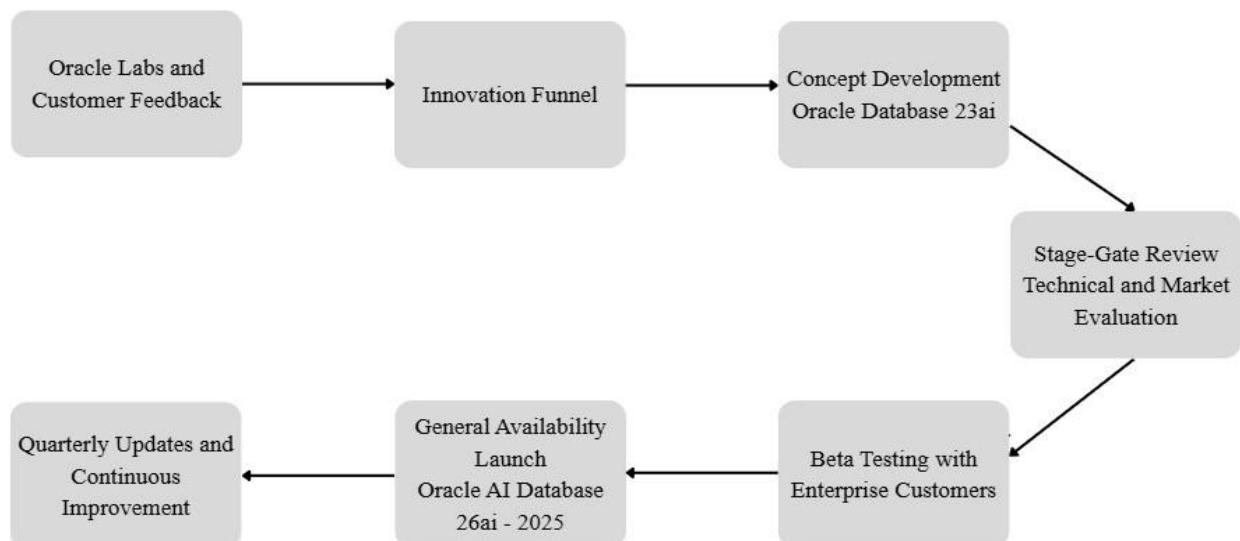


Figure 1. Oracle AI Database 26ai: Innovation Process

In Oracle's case, the path from creativity to innovation can be understood as a gradual movement

from early technical ideas to commercial release. The process starts with insights from Oracle Labs and enterprise customer requirements, then moves into an innovation funnel where ideas are screened for technical feasibility, market relevance, and strategic fit. The reference to Oracle Database 23ai represents the earlier product foundation where Oracle had already started building AI-related database capabilities. Oracle AI Database 26ai can then be seen as the later and broader commercialization of that direction, where those capabilities are expanded, validated, launched, and continuously improved after release.

This sequence also explains why the process should not be viewed as a simple one-step invention. Oracle first had to identify the enterprise problem, then turn the idea into concrete product capabilities, evaluate whether the project fit its long-term platform strategy, develop the product across technical workstreams, validate it with enterprise requirements, launch it commercially, and continue improving it after release. In other words, the value of 26ai comes not only from the original idea of putting AI inside the database, but from Oracle's ability to convert that idea into a scalable, secure, and commercially usable enterprise platform.

There's an important distinction the course makes between creativity and innovation: creativity is coming up with a new idea, but innovation only happens when that idea gets implemented and creates value (Schilling, 2023). Oracle's case illustrates this well. The creative insight behind 26ai, that AI should run inside the database rather than outside it, is relatively straightforward. The hard part is everything that comes after: engineering it into a production-grade product, getting it certified for enterprise security requirements, pricing it appropriately, and rolling it out across a global customer base.

2.2.1. How the idea is implemented commercially

For Oracle AI Database 26ai specifically, the process follows the course's creativity-to-innovation funnel. The idea started with recognizing a real enterprise problem: AI adoption is hard when sensitive business data is locked inside traditional databases, and moving that data to external AI systems creates security and compliance risks. Oracle's answer was to bring the AI to the data instead. From there, the team had to define what that actually meant in product terms (AI Vector Search, in-database agents, Autonomous AI Lakehouse), validate it with enterprise customers, build it across multiple engineering workstreams, and launch it globally through Oracle channels and the OCI ecosystem.

What's notable about Oracle's approach is the tension between speed and caution. Because Oracle's customers run mission-critical workloads, banking systems, hospital records, government data,

they can't treat 26ai like an experimental product. Security reviews, compatibility testing, and regulatory compliance all have to pass before anything ships. At the same time, the AI market moves fast enough that Oracle can't afford to run purely sequential development. The practical solution is a hybrid model: Stage-Gate reviews at major decision points, but parallel Agile sprints across engineering teams between those gates (Cooper, 1990).

Table 2. Creativity-to-Innovation Process Stages

Stage	Oracle activity	Academic interpretation
Idea generation	Recognize demand for AI- native enterprise data management.	Creativity and opportunity recognition.
Concept development	Define AI Vector Search, Autonomous AI Lakehouse, and in-database agent capabilities.	Transform broad idea into product concepts.
Project selection	Prioritize 26ai as a platform project with strategic value beyond immediate cash flow.	Screening questions and portfolio logic.
Development	Combine database engineering, cloud engineering, security, and AI capabilities.	Partly parallel development and cross-functional execution.
Customer/supplier validation	Use enterprise feedback and hardware/cloud partner input.	Lead-user involvement and supplier participation.
Commercial launch	Release 26ai and connect it to OCI/multicloud deployment.	Commercialization and diffusion.
Post-launch improvement	Deliver updates and expand ecosystem integration.	Continuous learning loop.

2.3. Types and patterns of innovation at Oracle

Schilling's framework asks us to classify innovation across four dimensions: product versus process, radical versus incremental, competence-enhancing versus competence-destroying, and component versus architectural (Schilling, 2023). Oracle AI Database 26ai doesn't fit neatly into just one category, the answer depends on which lens you apply.

2.3.1. Why the classification is mixed rather than single category

At the feature level, AI Vector Search is a component innovation, a new capability added to an

existing system without fundamentally changing the overall architecture. But zoom out to the platform level, and 26ai plus OCI together look more like architectural innovation: they reconfigure how databases, AI models, application tools, security systems, and cloud infrastructure interact with each other. For Oracle as a company, it's competence-enhancing, it builds directly on what Oracle already knows about RDBMS, enterprise security, and large-scale data management. For smaller players in the vector database space, though, it could be competence-destroying, because it bundles capabilities they built standalone products around.

Table 3. Types and Patterns of Innovation

Dimension	Classification for Oracle AI Database 26ai / OCI	Reasoning
Product vs. process	Primarily product innovation; also enabled by process innovation.	26ai is a new/updated service offering; OCI multicloud delivery and cloud operations improve delivery processes (Schilling, 2023).
Radical vs. incremental	Incremental for Oracle; more radical for some customers and competitors.	It builds on Oracle Database architecture but changes the role of the database in AI applications.
Competence-enhancing vs. competence-destroying	Competence-enhancing for Oracle; potentially competence-destroying for narrow vector- database rivals.	It extends Oracle RDBMS, security, and enterprise data capabilities into AI workloads.
Component vs. architectural	AI Vector Search is component innovation; the AI database + OCI + multicloud model is architectural innovation.	A feature is added to the engine, but the broader system reconfigures how data, AI models, applications, and clouds interact.

2.4. Technology S-curves and technology cycles

The S-curve framework is useful for understanding Oracle's timing (Schilling, 2023). Traditional relational databases sit at the mature phase of their S-curve, most of the fundamental performance gains have already been captured. Generative AI, vector search, and agentic computing are still in the steeper, early-growth phase. Oracle's strategy is to attach the new AI S-curve to its existing installed base, which theoretically speeds up adoption because enterprise customers don't need to

rebuild their entire data architecture from scratch.

That said, S-curves don't always bend the way companies expect. Knowing that a technology is valuable and actually adopting it are two different things. Many enterprise customers already understand the potential of AI Vector Search or Autonomous AI Lakehouse, but deploying them requires data governance policies, security reviews, budget approvals, and retraining, none of which happens quickly. Oracle's best advantage here is that it can reduce those friction points by keeping AI capabilities inside a platform customers already trust and manage. But it also faces a real risk: if cloud-native competitors become the default choice for new AI-data workloads before Oracle's installed base fully migrates, that advantage erodes.

PART III: FORMULATING TECHNOLOGICAL INNOVATION STRATEGY

3.1. Product overview and project-selection logic

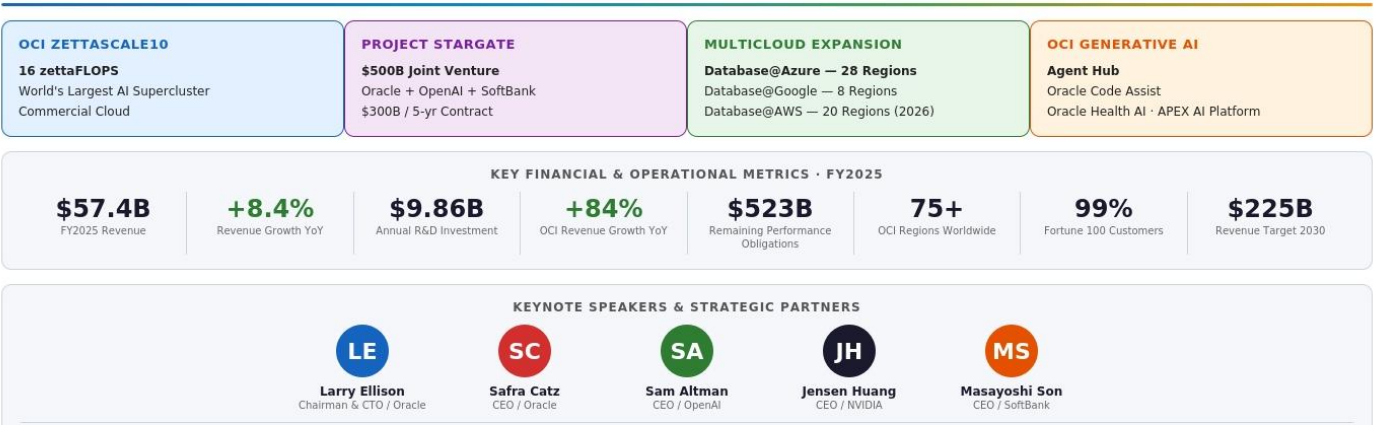


Figure 2. Oracle AI World 2025: Key Announcements and Strategic Metrics

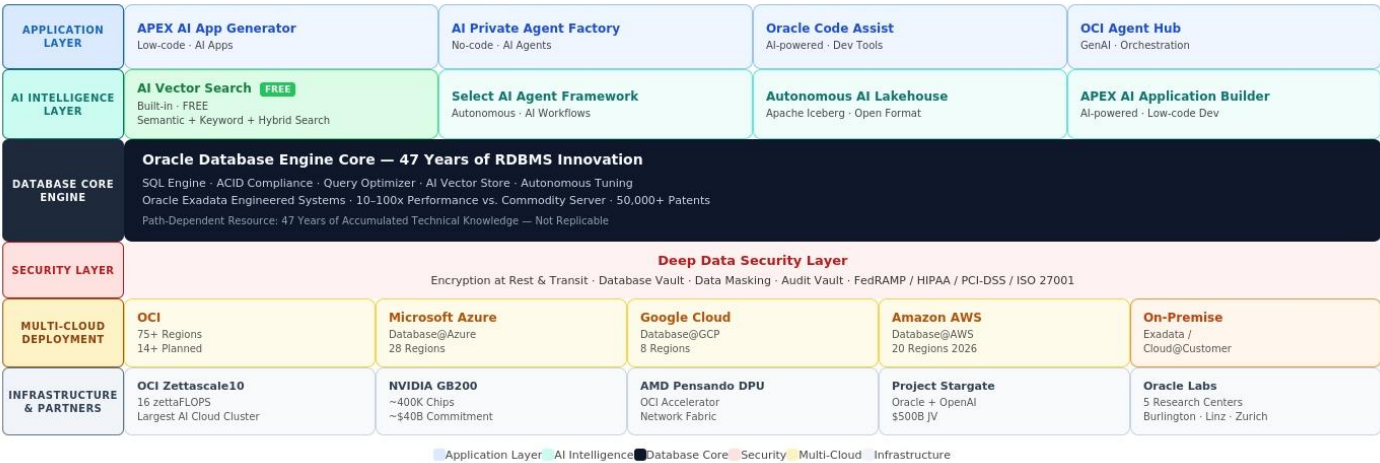


Figure 3. Oracle AI Database 26ai: Architecture Overview

Oracle AI Database 26ai is the focal product for this analysis. Launched in October 2025, it embeds

AI into the core database layer, with AI Vector Search included at no additional charge and with support for in-database AI agents, autonomous management, and AI Lakehouse capabilities. Instead of requiring enterprises to move sensitive data into a separate AI platform, Oracle's logic is to bring AI capabilities directly to governed enterprise data. This is strategically important because it links the new AI wave to Oracle's long-established strength in mission-critical database systems.

The AI World summary and the architecture overview make the project-selection logic more concrete. Oracle is clearly not treating 26ai as a small database update. The product sits inside a much broader AI infrastructure and platform strategy, supported by FY2025 revenue of US\$57.4 billion, annual R&D investment of US\$9.86 billion, OCI revenue growth of +84% year over year, remaining performance obligations of US\$523 billion, more than 75 OCI regions worldwide, 99% penetration among Fortune 100 customers, and a long-term revenue target of US\$225 billion by 2030. These numbers help explain why 26ai should be evaluated as a platform project with future option value, rather than only through short-term cash-flow logic.

The architecture overview also shows why the project is strategically feasible. AI capabilities are built above Oracle's database core, which rests on 47 years of RDBMS innovation. Around that core, Oracle connects AI Vector Search, Select AI Agent Framework, Autonomous AI Lakehouse, APEX AI App Generator, Oracle Code Assist, and OCI Agent Hub with a deep data security layer and a multicloud deployment layer. The point is not only that Oracle is launching another AI feature. The stronger argument is that Oracle is trying to make the database layer itself the foundation for enterprise AI workloads across applications, intelligence, security, and cloud infrastructure.

From a project-selection standpoint, 26ai satisfies the main screening questions discussed in innovation strategy theory. First, it leverages Oracle's core competencies in database engineering, security, enterprise relationships, and cloud infrastructure. Second, it protects Oracle's installed base by allowing existing enterprise customers to adopt AI without abandoning Oracle Database. Third, it creates future options because AI Vector Search, in-database agents, Autonomous AI Lakehouse, and OCI infrastructure can support many future products and services. Fourth, it opens new markets in enterprise AI, AI-data platforms, agentic applications, and multicloud AI deployment. For this reason, Oracle AI Database 26ai is best understood as a platform project with breakthrough potential, not merely an incremental database release (Schilling, 2023).

3.2. External analysis: Porter's Five Forces

The relevant competitive arena is the enterprise cloud database and AI-data infrastructure market.

Porter's Five Forces are useful because they evaluate industry attractiveness rather than only product features (Porter, 1980).

Table 4. Porter's Five Forces Assessment

Force	Assessment	Evidence and strategic interpretation
Threat of new entrants	Low to medium	Enterprise cloud database requires capital, trust, compliance, uptime, security, and ecosystem credibility. Startups can enter narrow vector-search niches, but competing with Oracle across mission-critical enterprise workloads is much harder (Porter, 1980).
Supplier power	Medium to high	AI infrastructure depends on GPUs, networking, power, and data-center construction. NVIDIA and other advanced- chip suppliers have strong bargaining power. Oracle must diversify hardware and infrastructure relationships.
Buyer power	Medium	Legacy enterprise customers face high switching costs, but AI-first customers can choose AWS, Azure, Google Cloud, Snowflake, Databricks, MongoDB, Pinecone, or open- source alternatives.
Threat of substitutes	Medium to high	Substitutes include NoSQL, specialized vector databases, lakehouse platforms, data warehouses, and PostgreSQL + pgvector. Oracle responds by embedding AI Vector Search and lakehouse capabilities inside the database platform.
Industry rivalry	Very high	AWS, Microsoft Azure, Google Cloud, Snowflake, Databricks, MongoDB, and open-source ecosystems compete for AI- data workloads. Oracle cannot win only on price; differentiation and multicloud strategy are more appropriate.

Reading across these five forces, Oracle's best defensive position isn't competing on price, that's a losing game against AWS or Azure. Instead, Oracle's advantage is differentiation: secure, governed enterprise AI sitting on top of data that customers already store in Oracle systems, and increasingly deployable wherever those customers prefer to run their workloads. The multicloud strategy is particularly smart because it turns rivals into distribution channels. LLM providers and

application developers who integrate with Oracle's platform also reinforce this position, making the ecosystem more valuable the more participants join it.

3.3. Internal analysis: Porter's Value Chain

PRIMARY ACTIVITIES				
Inbound Logistics	Operations	Outbound Logistics	Marketing & Sales	Service
✓ Strengths NVIDIA/AMD contracts 75+ OCI regions ✗ Weaknesses GPU lead time 6-12 months delay	✓ Strengths Exadata 10-100x perf OCI +84% YoY Oracle Labs x5 ✗ Weaknesses CAPEX \$25B/year Quarterly release cycle	✓ Strengths Cloud@Customer +104% YoY growth Multicloud (3 providers) ✗ Weaknesses Coverage gaps: Africa Eastern Europe	✓ Strengths RPO \$523B +438% 99% Fortune 100 AI World 30K attendees ✗ Weaknesses Marketing spend \$296M Weak cloud brand	✓ Strengths MOS 135K staff 175 countries \$8.7B service revenue ✗ Weaknesses Poor SMB experience Lack of self-service
SUPPORT ACTIVITIES				
Firm Infrastructure	Human Resource Mgmt	Technology Development (R&D)	Procurement	
✓ Strengths Ellison ~40% stake Long-term strategic focus ✗ Weaknesses Oracle-way culture Excessive centralization	✓ Strengths 162K employees Oracle University Follow-the-sun 24/7 ✗ Weaknesses Hard to recruit AI talent OpenAI/Google competition	✓ Strengths R&D \$9.86B (35% OPEX) 50K patents Oracle Labs >\$90B accumulated since FY2012 ✗ Weaknesses Quarterly release cycle Slower than AWS continuous	✓ Strengths Scale buying power NVIDIA/AMD/Intel deals ✗ Weaknesses Dependent on 1-2 GPU suppliers Lack of custom silicon	

Figure 4. Value Chain Analysis: Oracle AI Database 26ai (Porter, 1985)

Porter's Value Chain breaks the firm's activities into primary activities , those directly involved in creating and delivering the product , and support activities that enable those primary functions (Porter, 1985). For Oracle, each activity has clear strengths tied to its size and history, but also specific risks in the AI-era context.

Table 5. Value Chain: Primary and Support Activities

Activity	Strength for Oracle	Weakness / risk
Inbound logistics	Oracle's AI infrastructure benefits from NVIDIA/AMD hardware relationships and a global cloud footprint of 75+ OCI regions. These inputs support the compute capacity required for enterprise AI workloads.	AI infrastructure remains exposed to GPU supply constraints, long hardware lead times, energy availability, and semiconductor geopolitics. Figure 4 specifically points to GPU lead-time pressure of 6-12 months.
Operations	Oracle has deep database engineering capabilities, Exadata infrastructure, OCI cloud operations, and Oracle Labs. Figure 4 links this to Exadata performance advantages, OCI's strong growth, and Oracle's ability to operate mission-critical systems at global scale.	Scaling AI infrastructure is capital intensive. Figure 4 highlights high CAPEX requirements, and the need for frequent release cycles can create pressure on reliability, testing, and cost control.

Outbound logistics	Oracle can deliver 26ai through OCI, Database@Azure, Google Cloud, AWS partnerships, Cloud@Customer, and on-premises Exadata. This strengthens adoption because customers can run Oracle technologies close to existing workloads.	The main delivery risk is not only partner-cloud pricing but also geographic coverage gaps. Figure 4 highlights weaker coverage in Africa and Eastern Europe, which can limit diffusion in some markets.
Marketing and sales	Oracle has strong enterprise sales relationships, large contracted demand, and a trusted brand among mission-critical customers. Figure 2 and Figure 4 highlight US\$523 billion in remaining performance obligations, 99% Fortune 100 penetration, and strong AI World visibility.	Oracle still has weaker developer and startup mindshare than cloud-native rivals. Even with strong enterprise demand, the company must improve AI developer experience and self-service adoption.
Service	Oracle's global support organization and enterprise account management help reduce adoption risk for large customers. Figure 4 connects service strength to My Oracle Support, worldwide coverage, and service revenue.	The support model can be complex and expensive for smaller firms. Weak self-service experience can slow adoption among developers, startups, and smaller AI-native companies.
Firm infrastructure	Oracle benefits from stable leadership, financial scale, and long-term strategic control. Figure 4 notes Larry Ellison's large ownership stake, which can support long-term commitment to database and AI infrastructure strategy.	Founder influence and centralization can also become governance risks if decision-making becomes too concentrated or if the company over-invests in infrastructure before demand fully materializes.
Human resources	Oracle's large global workforce, Oracle University, and follow-the-sun operating model support enterprise	Competition for elite AI engineers is intense, especially against OpenAI, Google, Microsoft, AWS,

	implementation, training, and customer success.	and AI-native startups.
Technology development (R&D)	Oracle's FY2025 R&D expense of US\$9.86 billion, Oracle Labs, long database history, and large patent base support sustained innovation. Figure 3 and Figure 4 reinforce this through the claims of 47 years of RDBMS innovation, 50K+ patents, and accumulated R&D investment.	The same formalization that protects database reliability may slow AI-era experimentation. Oracle must avoid allowing core competencies to become core rigidities.
Procurement	Oracle's purchasing scale and strategic hardware relationships strengthen its ability to secure GPUs, networking, and data-center capacity.	Oracle still depends heavily on external chip suppliers and lacks full control over proprietary AI silicon, which increases supplier power and infrastructure risk.

The value-chain analysis shows that Oracle's strongest advantage lies in the combination of technology development, operations, outbound logistics, and enterprise sales. In technology development, Oracle's long history in database innovation, large R&D investment, Oracle Labs, and extensive patent base create a strong foundation for AI Database 26ai. This supports the argument that 26ai is not an isolated AI feature, but an extension of Oracle's accumulated database and infrastructure capabilities. In operations, Oracle's Exadata, OCI, and database engineering capabilities allow the company to deliver AI functions inside mission-critical data systems, which is difficult for smaller AI-native firms to replicate.

The analysis also shows that Oracle's outbound logistics and marketing capabilities are strategically important. Its multicloud deployment model through OCI, Azure, Google Cloud, AWS, Cloud@Customer, and on-premises Exadata helps reduce customer adoption barriers because enterprises can run Oracle technologies close to their existing workloads. At the same time, Oracle's large enterprise customer base, remaining performance obligations, and Fortune 100 penetration strengthen its commercial position. These factors make AI Database 26ai more credible as a platform project because Oracle already has the customer relationships and distribution channels needed to diffuse the innovation.

However, the value chain also reveals several weaknesses. Oracle remains exposed to GPU supply constraints, high infrastructure investment, energy availability, and dependence on external chip suppliers. Its weaker developer and startup mindshare also creates a challenge because the AI market is increasingly influenced by developer communities and cloud-native ecosystems. In addition, centralization and formalization help Oracle maintain reliability and compliance, but they can slow experimentation in AI tooling, interface design, and developer experience.

In short, Oracle's value chain supports the strategic logic of AI Database 26ai, but it also shows the main execution risks. Oracle is strongest where enterprise trust, database reliability, R&D depth, and multicloud reach matter most. Its main challenge is to maintain these enterprise strengths while improving speed, developer adoption, infrastructure flexibility, and AI-era experimentation.

3.4. Core competencies and VRIO analysis

Core competencies are not simply individual products or isolated features. Following Prahalad and Hamel, a capability should be treated as a core competency only when it contributes clearly to customer value, can be used across multiple markets, and is difficult for competitors to imitate (Prahalad & Hamel, 1990). Therefore, Oracle's analysis should begin by identifying the competency bundle behind Oracle AI Database 26ai rather than treating 26ai as only a single product feature.

Oracle's first core competency is database engineering and reliability. This capability is valuable because it allows Oracle to embed AI Vector Search, in-database agents, autonomous tuning, compatibility, and mission-critical reliability directly inside the database engine. It is rare and difficult to imitate because it depends on 47 years of RDBMS experience, accumulated technical knowledge, customer trust, and a large patent base. The architecture and value-chain evidence both point to the same conclusion: Oracle's advantage is rooted in long-term database knowledge, Oracle Labs, and intellectual-property depth. Under VRIO logic, this gives Oracle a sustained advantage when enterprise customers require AI on governed data rather than detached experimental AI tools (Barney, 1991; Prahalad & Hamel, 1990).

The second competency is enterprise trust and regulated-industry knowledge. Oracle has long served banks, healthcare organizations, telecom firms, governments, and other customers that require uptime, security, auditability, and compliance. This capability is valuable because it reduces adoption barriers for AI in sensitive environments. It is partly rare, but not completely inimitable, because major hyperscalers can also offer compliance, cloud credits, and enterprise support. Therefore, this competency creates a sustained-to-temporary advantage: strong today, but

requiring continuous renewal.

The third competency is security and data governance. The deep data security layer, including encryption, auditability, and compliance-oriented controls, strengthens Oracle's value proposition because many enterprises do not want to move sensitive data into external AI systems. Oracle's strategy of bringing AI to governed data directly supports high-risk use cases where privacy, access control, and audit trails matter. Under VRIO logic, this capability is valuable, rare in its database-integrated form, and difficult to imitate when combined with Oracle's installed base and enterprise routines.

The fourth competency is OCI, Exadata, and multicloud deployment. Oracle can connect AI Database 26ai with OCI, Database@Azure, Google Cloud, AWS, Cloud@Customer, and on-premises infrastructure. This gives customers flexibility and turns cloud rivals into distribution partners. However, the advantage is not risk-free: hyperscalers can improve their own databases, vector search, and AI-data platforms. Therefore, this capability should be treated as a new sustained-to-temporary advantage that depends on execution quality and ecosystem adoption.

The final competency is AI-data integration. Individual AI features such as vector search or agents can eventually be copied, but the integrated bundle is harder to replicate: database reliability, security governance, AI search, lakehouse support, agents, analytics, developer tools, and multicloud delivery. This combination is the real source of Oracle's strategic advantage. The VRIO conclusion is therefore that Oracle's moat does not come from AI Vector Search alone. It comes from the bundle of competencies that allows Oracle to transform 26ai from a technical feature into a broader enterprise AI-data platform (Barney, 1991; Leonard-Barton, 1992; Prahalad & Hamel, 1990).

3.5. Collaboration strategy and innovation protection

A key strategic question for any innovation project is whether to develop it solo or collaborate, and if collaborating, what kind of agreement makes most sense. Going solo keeps proprietary knowledge under control and protects competitive advantage, but it's slow and resource-intensive. Collaboration trades some control for speed, access to scarce capabilities, and risk-sharing. Oracle uses both approaches simultaneously, depending on which layer of the product is involved.

Table 6. Collaboration Strategy and Innovation Protection

Product / activity layer	Mode used by Oracle	Reasoning
Oracle AI	Solo internal	The database engine is the strategic asset; internal

Database engine core	development	development protects trade secrets, reliability, roadmap control, and accumulated database competencies (Schilling, 2023).
AI model and developer integration layer	Selective collaboration / open integration	External LLMs and developer tools improve adoption; the integration layer should remain open enough for ecosystem diffusion.
OCI Zettascale10 and AI infrastructure	Strategic alliances	Oracle needs specialized GPUs, networking, power, and data-center capabilities that are too slow or costly to build alone.
Stargate AI infrastructure	Joint venture	Oracle contributes OCI infrastructure while OpenAI provides demand and SoftBank contributes capital and infrastructure participation (OpenAI, 2025).
Oracle Database@Azure / Google Cloud / AWS	Strategic alliance / coopetition	Oracle turns cloud rivals into distribution partners so customers can run Oracle databases near existing cloud workloads.

On the protection side, Oracle applies what Schilling calls selective appropriability , protecting different parts of the product to different degrees (Schilling, 2023). The database engine core is kept highly proprietary (trade secrets, patents, restricted access) because it's the primary source of long-term differentiation. AI Vector Search, by contrast, is included free to drive adoption and prevent substitute platforms from capturing that workload. APIs and developer tools are partially open to encourage third-party integration, while brands (Oracle, Java, OCI, APEX) are trademark-protected to maintain identity in a crowded AI market.

PART IV: IMPLEMENTING TECHNOLOGICAL INNOVATION STRATEGY

4.1. New Product Development process for Oracle AI Database 26ai

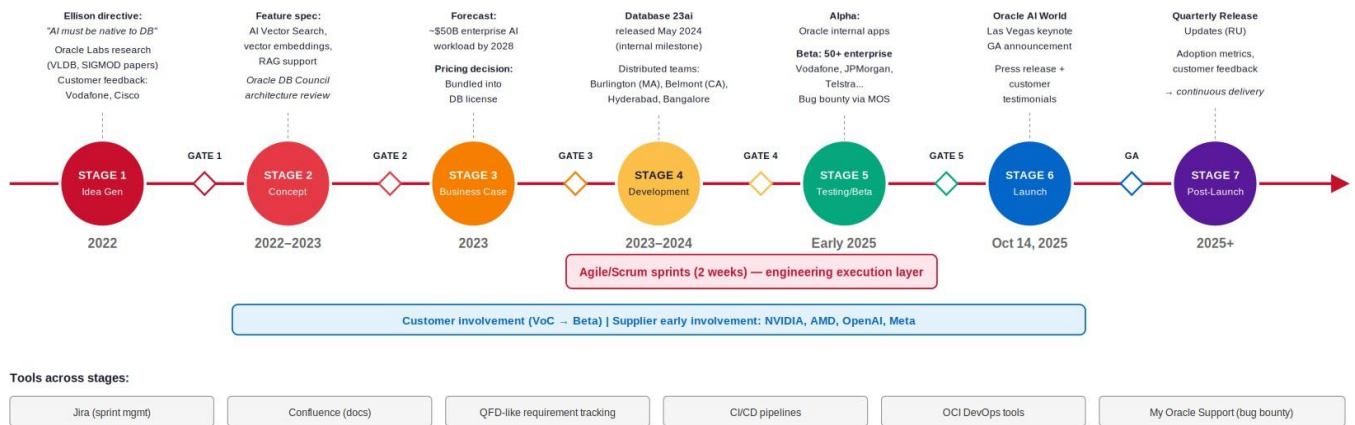


Figure 5. Stage-Gate and Agile Hybrid NPD Process

The timeline below should be understood as a theory-based reconstruction of the development logic, not as Oracle's disclosed internal project plan. It is included to connect the course's NPD framework with the public information available about Oracle AI Database 26ai.

Oracle has publicly announced the product, its AI capabilities, and its cloud infrastructure strategy, but it has not released a detailed internal development workflow for 26ai. For that reason, the discussion below avoids claiming access to Oracle's confidential process and instead uses NPD theory to interpret how a product of this scale would need to be managed.

From an NPD theory perspective, Oracle AI Database 26ai can be interpreted as fitting a hybrid Stage-Gate and Agile logic. A purely sequential process would be too slow because AI technologies, enterprise customer expectations, and competitive moves from AWS, Microsoft Azure, Google Cloud, Snowflake, Databricks, and open-source ecosystems change rapidly. However, a fully experimental startup-style process would also be inappropriate because Oracle's customers use its databases for mission-critical workloads that require reliability, backward compatibility, cybersecurity, data governance, and regulatory compliance.

The table below translates that interpretation into seven practical stages. It connects the development logic of Oracle AI Database 26ai with the course concepts of market fit, concept screening, partly parallel development, commercialization, and post-launch learning.

Table 7. NPD Process Stages for Oracle AI Database 26ai

NPD stage	Oracle activity	Theory applied
1. Opportunity recognition	Identify enterprise demand for secure AI on governed data.	Market/customer fit (Cooper, 1990).
2. Concept development	Specify AI Vector Search, in-database agents, lakehouse support, autonomous tuning, and	Customer requirements and concept screening.

	multicloud deployment.	
3. Business case and portfolio fit	Treat 26ai as a strategic platform project, not merely a derivative release.	Project selection, screening questions, and strategic option value.
4. Development	Run database, OCI, security, AI-data services, and application capabilities in overlapping workstreams.	Partly parallel development / Agile execution.
5. Customer and supplier involvement	Use enterprise feedback and inputs from AI hardware and cloud ecosystem partners.	Lead-user and supplier involvement.
6. Launch	Release through Oracle channels and connect with AI World messaging.	Commercialization and diffusion.
7. Post-launch improvement	Deliver updates, expand integrations, improve developer experience, and incorporate customer feedback.	Closed-loop learning and continuous improvement.

The first stage is opportunity recognition. Oracle identified a clear enterprise problem: organizations want to use AI on business data, but moving sensitive data into external AI systems creates security, governance, and compliance risks. This explains the core strategic idea behind 26ai: bring AI to the data rather than forcing data to move to separate AI tools. The second stage is concept development, where this broad idea becomes concrete product capabilities such as AI Vector Search, in-database agents, Autonomous AI Lakehouse, autonomous tuning, and multicloud deployment.

The third stage is business-case and portfolio evaluation. 26ai is not only a database feature; it supports Oracle's broader AI-data platform strategy. The project is justified because it protects the installed database base, creates future options in enterprise AI, strengthens OCI demand, and supports multicloud distribution. The fourth stage is development. In theory, this requires overlapping workstreams across database engineering, OCI infrastructure, security, AI-data services, application tools, and customer-facing integration. This is where Agile-style parallel

execution is useful, because different technical streams must progress at the same time.

The fifth stage is customer and supplier involvement. Enterprise customer requirements shape the need for security, compliance, high availability, and deployment flexibility, while suppliers and partners such as NVIDIA, AMD, OpenAI, Microsoft Azure, Google Cloud, AWS, and SoftBank contribute infrastructure, model ecosystems, or distribution reach. The sixth stage is launch, represented by Oracle's public AI World messaging and the October 2025 26ai announcement. The seventh stage is post-launch improvement, where Oracle must continue expanding integrations, improving developer experience, strengthening security, and responding to customer feedback.

The main point is not that Oracle has publicly disclosed every internal stage or tool. What matters for this analysis is the balance between speed and control. Stage-Gate logic helps with technical feasibility, security readiness, customer value, and commercial-readiness decisions, while Agile logic supports parallel product engineering and faster iteration. For a mission-critical AI database product, neither pure sequential development nor uncontrolled experimentation would be sufficient (Cooper, 1990).

4.2. Implementation team structure for Oracle AI Database 26ai

Oracle has not publicly disclosed a confidential engineering-squad chart for Oracle AI Database 26ai. Therefore, the accurate way to analyze the implementation team is not to invent an internal project chart, but to map Oracle's official public leadership roles to the functions required to deliver 26ai. Public Oracle sources show a structure that connects executive technology direction, Oracle Database Technologies, AI Data Platform, OCI, security/legal governance, field operations, and customer success (Oracle Corporation, n.d.-a, n.d.-b, n.d.-c, n.d.-d, 2025d).

At the public level, this structure resembles a heavyweight cross-functional implementation model: product ownership remains close to Oracle's core database and cloud organizations, while specialist functions support security, compliance, go-to-market execution, and customer adoption. This matters because 26ai is not a standalone interface project; it depends on the database engine, AI-data services, cloud infrastructure, enterprise security, and multicloud delivery working together.

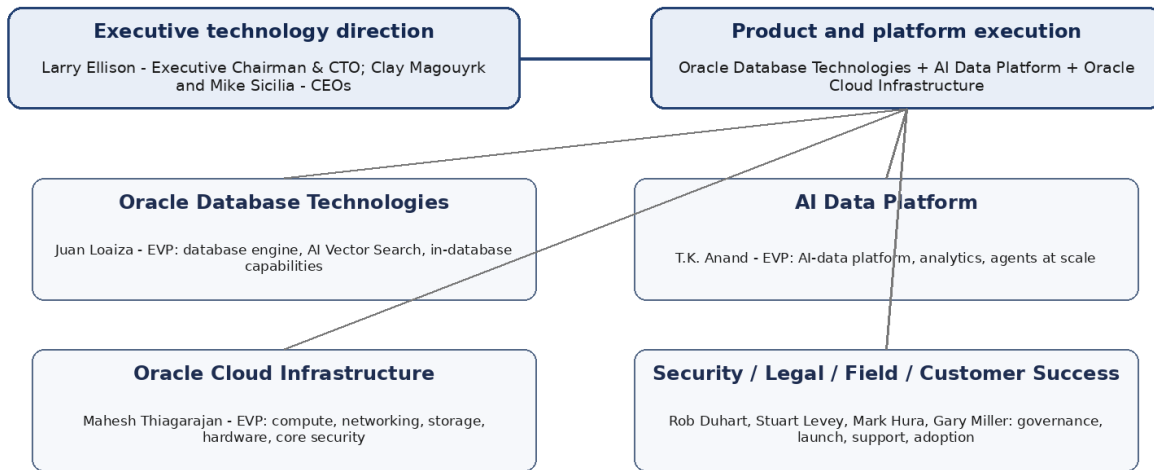


Figure 6. Publicly Evidenced Oracle Implementation Structure

Table 8. Oracle Implementation Structure for AI Database 26ai

Public Oracle role / function	Evidence	Implementation relevance to 26ai
Executive technology direction	Lawrence J. Ellison is Oracle Executive Chairman and CTO; Oracle also promoted Clay Magouyrk and Mike Sicilia to CEO roles in 2025 (Oracle Corporation, n.d.-a, 2025d).	Sets strategic direction and coordinates the enterprise AI, database, cloud, and industry-application agenda.
Oracle Database Technologies	Juan Loaiza is EVP, Oracle Database Technologies, and Oracle describes him as responsible for driving innovation and development of Oracle flagship database products (Oracle Corporation, n.d.-b).	Owns the core database layer where AI Vector Search, in-database AI capabilities, autonomy, and compatibility must be engineered.
Healthcare and AI Data Platform	T.K. Anand is EVP, Healthcare and AI Data Platform; Oracle states that his AI Data Platform team builds foundational data and	Connects database innovation with AI-data services, analytics, agent orchestration, and enterprise use cases.

	AI infrastructure and orchestrates AI agents at scale (Oracle Corporation, n.d.-c).	
Oracle Cloud Infrastructure	Mahesh Thiagarajan is EVP, OCI, leading development and operation of compute, networking, storage, hardware, data center, supply chain, developer, and core security services (Oracle Corporation, n.d.-d).	Provides the infrastructure base for AI workloads, distributed cloud deployment, and high-performance enterprise cloud operations.
Security, legal, field operations, and customer success	Oracle lists Rob Duhart as Chief Security Officer, Stuart Levey as Chief Legal Officer, Mark Hura as President of Global Field Operations, and Gary Miller as Customer Success Officer (Oracle Corporation, n.d.-a).	Ensures security governance, legal/compliance readiness, commercial launch, customer onboarding, and post-launch adoption support.

The strength of this structure is that each implementation function is tied to an existing Oracle capability rather than being treated as a temporary project group. Database Technologies protects the product core; OCI supplies infrastructure scale; AI Data Platform links the product to agent and analytics use cases; and security, legal, field, and customer-success functions reduce enterprise adoption risk. The limitation is that Oracle's public materials do not reveal the lower-level engineering squad structure, so our team decided describe the publicly evidenced implementation structure rather than claim access to confidential internal team design.

4.3. Product Prototype: Oracle AI Database 26ai Interface

The prototype turns the strategy discussion into a possible user-facing product concept. It is not an official Oracle interface or product screenshot. It was designed by the authors for academic illustration, using Oracle's public product announcements, the architecture logic discussed earlier, and the strategic analysis developed throughout the report. Its purpose is to show how Oracle AI Database 26ai could be communicated to enterprise users as an AI-native data platform.

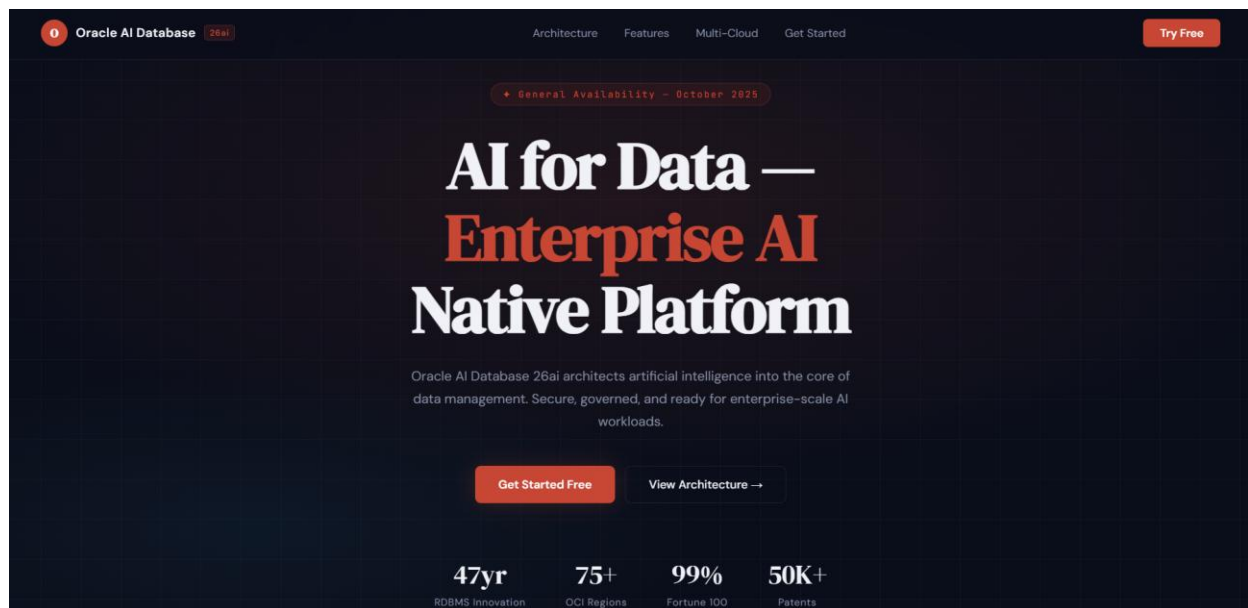


Figure 7. Prototype Landing Page: Oracle AI Database 26ai Enterprise AI-Native Platform

The landing page opens with Oracle AI Database 26ai positioned as an Enterprise AI Native Platform. That message reflects the central argument of the report: Oracle is not simply adding AI features to a database, but trying to reposition the database as the natural home for enterprise AI. The metrics shown on the screen, including 47 years of RDBMS experience, 75+ OCI regions, 99% Fortune 100 penetration, and a large patent base, help make Oracle's credibility visible to the user.

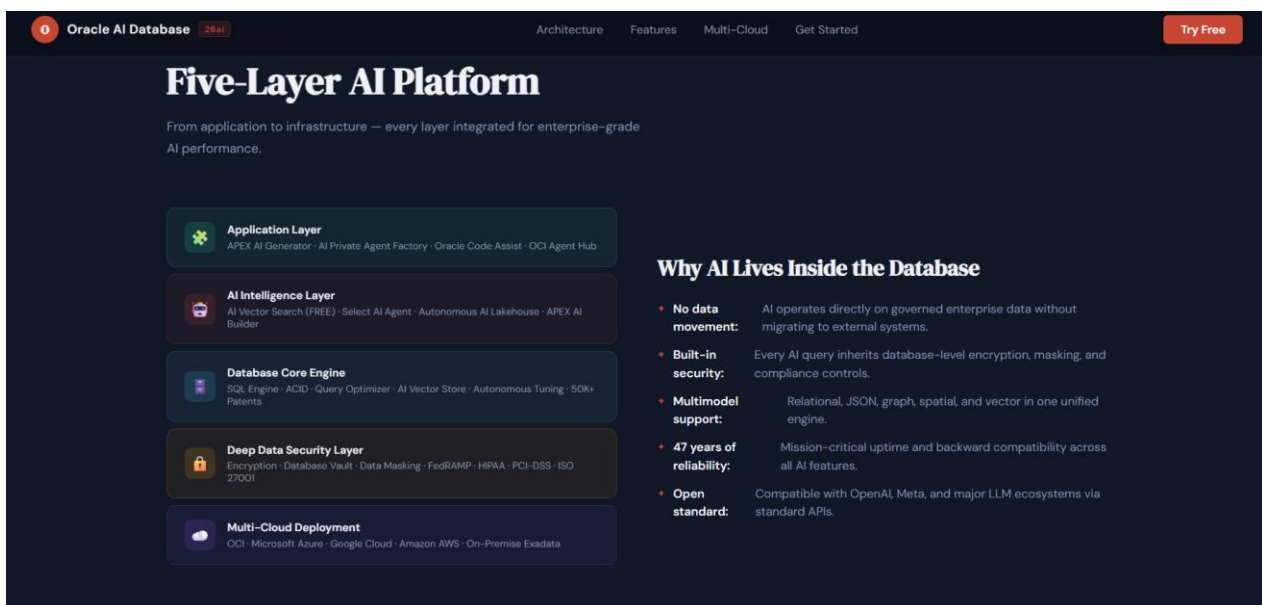


Figure 8. Prototype Architecture View: Five-Layer AI Platform

The architecture screen breaks the platform into five layers: application, AI intelligence, database core engine, deep data security, and multicloud deployment. This structure mirrors the earlier architecture analysis. The section on why AI should live inside the database also captures the main

strategic value proposition: less data movement, built-in security, multimodel support, long-term database reliability, and openness through standard interfaces. These points connect directly to the VRIO discussion because Oracle's advantage comes from the whole capability bundle, not from one isolated AI function.

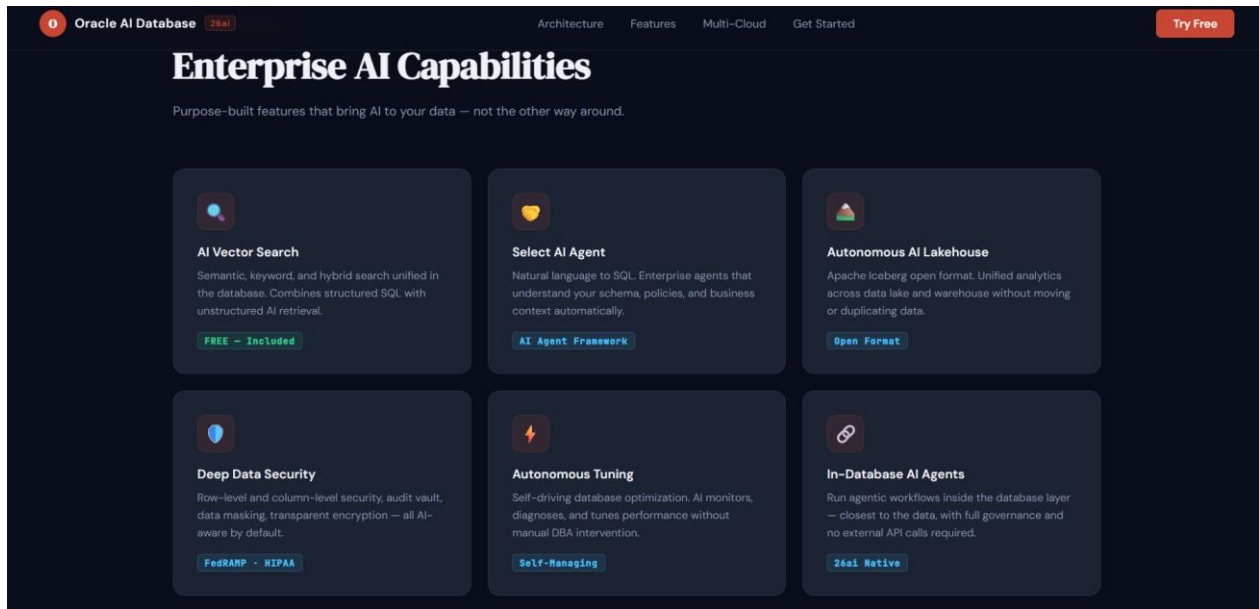


Figure 9. Prototype Feature View: Enterprise AI Capabilities

The capability screen makes the platform more concrete by showing six enterprise AI functions: AI Vector Search, Select AI Agent, Autonomous AI Lakehouse, Deep Data Security, Autonomous Tuning, and In-Database AI Agents. These cards support the innovation-type analysis in Part II. Some of them are component innovations, while the combined platform becomes closer to architectural innovation because it changes how data, AI, security, analytics, and cloud deployment work together.

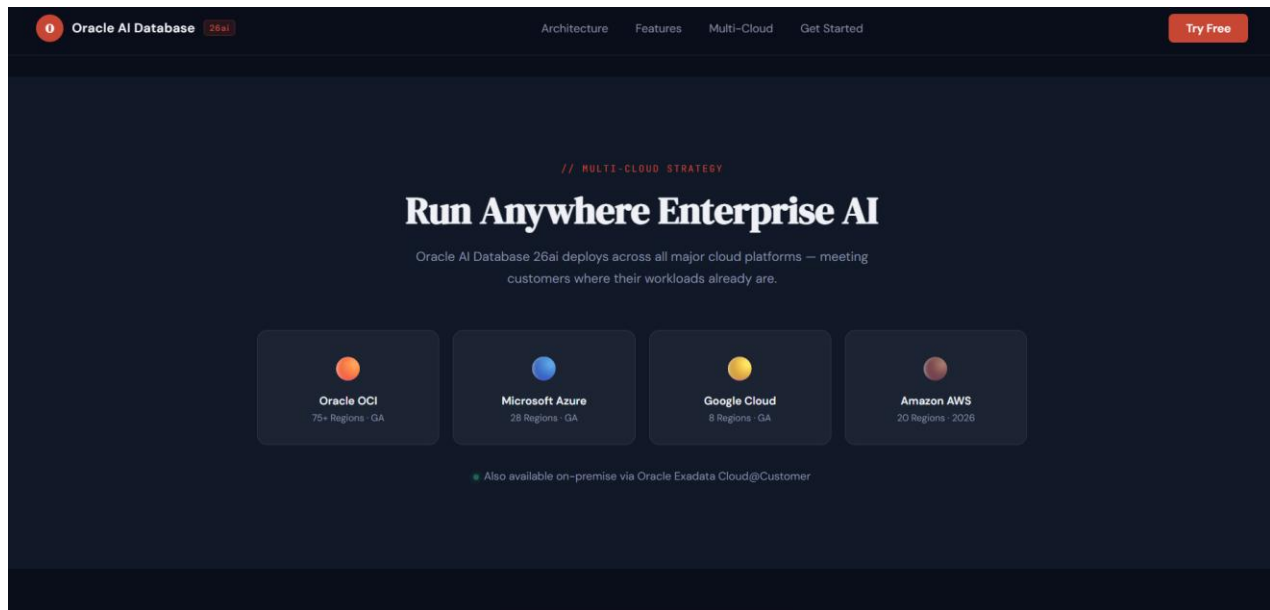


Figure 10. Prototype Multicloud View: Run Anywhere Enterprise AI

The multicloud screen emphasizes that the product concept is not tied to a single cloud. By presenting the message as Run Anywhere Enterprise AI, the prototype connects directly with Oracle's collaboration strategy. Database@Azure, Google Cloud, AWS, OCI, Cloud@Customer, and on-premises deployment all reduce adoption friction because customers can keep workloads close to their existing cloud and database environments. Strategically, this is how Oracle can turn competitors into distribution partners rather than treating every hyperscaler only as a rival.

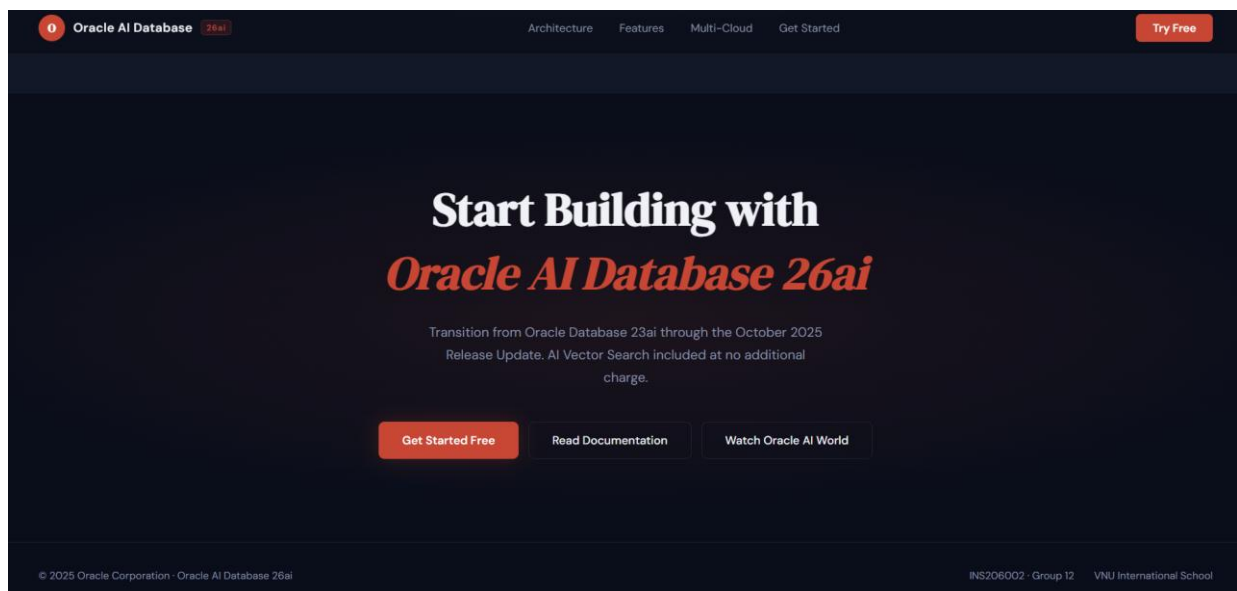


Figure 11. Prototype Call-to-Action View: Start Building with Oracle AI Database 26ai

The final screen links the prototype to commercialization. A useful product concept should not stop at technical architecture; it should also guide users toward trials, documentation, product learning, and adoption. This final view therefore connects back to the creativity-to-innovation

process: the idea creates value only when customers can understand it, test it, and integrate it into real enterprise workflows.

PART V: CONCLUSION

Oracle AI Database 26ai is strategically important because it shows how a mature technology company can use existing core competencies as the foundation for a new AI-era platform. The central finding of this report is that Oracle's advantage does not come from one isolated AI feature. It comes from the combination of database engineering, enterprise trust, security governance, OCI infrastructure, and multicloud distribution.

The analysis also shows why Oracle's implementation structure matters. A hybrid Stage-Gate and Agile NPD process is appropriate because 26ai must move fast enough for the AI market while still meeting enterprise expectations for uptime, compatibility, security, and compliance. Oracle's publicly evidenced structure works like a heavyweight cross-functional model because database technology, OCI infrastructure, AI Data Platform, security/legal governance, field operations, and customer success are connected around the same enterprise AI-data platform requirements.

Overall, Oracle's best strategic path is to protect the proprietary database core while keeping the surrounding AI and developer ecosystem open enough to encourage adoption. The company must still manage supplier dependence, hyperscaler rivalry, and the risk of bureaucratic inertia. However, if Oracle can maintain the balance between reliability and speed, Oracle AI Database 26ai can reinforce Oracle's role as a major enterprise AI-data platform rather than merely a traditional database vendor (Teece et al., 1997).

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